What momentum says about impact

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect stopping impulse to average force without claiming a complete damage forecast.

SUCCESS CRITERIA - I can...

1. Compute momentum change during a stop.
2. Compare average force for different stopping times.
3. Name information beyond momentum needed for damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “impact” and “momentum” to explain your answer. Sentence starters are optional.

impact: a collision or strike between objects

momentum: mass times velocity, including direction

interval: the time or range between two stated values

01 Warm-up: bring back an earlier idea

Two **3 kg** carts at **+5** and **-3 m/s** stick. **Find final velocity** and use kinetic energy to **explain why this** is not elastic.

Word help: kinetic energy = energy associated with motion

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p = mv$; $|F_{avg}| = |\Delta p|/\Delta t$.
Stopping force depends on stopping time as well as momentum. These quantities alone do not specify damage.

Worked reasoning: Two incoming objects have equal momentum **12 kg·m/s** but masses **2** and **4 kg**, so their speeds are **6** and **3 m/s**. Momentum alone does not name every possible consequence of stopping them.

Your turn: Draw equal momentum arrows and unequal velocity arrows.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



2 kg; 6 m/s 4 kg; 3 m/s

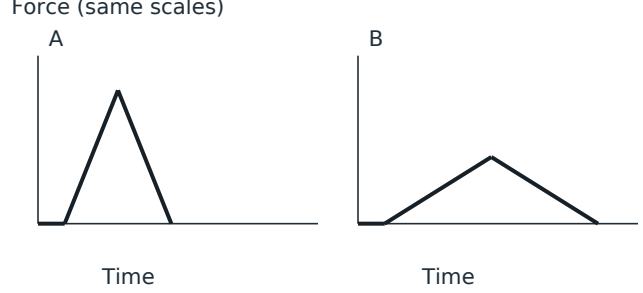
Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two equal-momentum objects stop in **0.1 s** and **1 s**. Which has greater average force magnitude, and by what factor?

Word help: magnitude = size of a quantity without its direction

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

04 Apply the physics

A hypothetical object of mass **10^6 kg** at **200 m/s** stops in **0.50 s**. **Find impulse** and average force magnitude. Then **compare a 5 s** stop.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Both ... . They differ in ... because ... .

Object / trial	Mass	Velocity	Momentum
A	___	___	___
B	___	___	___

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why is a single stopping-force number not a prediction of buildings damaged by an asteroid?

Sentence starter: I predict … . I used … to make this prediction because … .

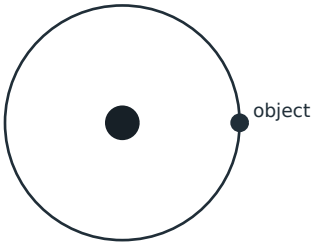
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An ideal circular orbit has GM=**576 m³/s²** and r=**16 m**. **Find speed**, inward acceleration and period; **explain how velocity changes**.
Label velocity and inward acceleration on the circular-path diagram.

Word help: model = a drawing, explanation or equation for a physical situation

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: 1 2 3

07 Exit check

Objects A and B have equal momentum. A has twice B's mass. **Compare speeds** and kinetic energies, showing why equal momentum need not mean equal energy.

Sentence starter: Both … . They differ in … because … .

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **0.5 kg** ball arrives at **+10 m/s** and either stops or rebounds at **-6 m/s** in **0.04 s**. **Compare impulse** and average force.

Sentence starter: Both … . They differ in … because … .

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3